Ekansh Bajpai

Robotics researcher: event-based vision, SLAM and humanoid robot learning

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Research interests

Event-based and neuromorphic vision (spiking neural networks) for visual odometry and SLAM; learning-based perception and control for agile robots, from imitation learning with vision-language-action models to whole-body control of humanoids.

Education

M.Sc. Electrical Systems Engineering 2026
Paderborn University, Germany
Master's thesis: *Exploring Neuromorphic Front-Ends for Event-Based SLAM* — **grade 1.0**
Supervisor: Dr. Alper Yegenoglu; examiners: Prof. Dr. Erdal Kayacan, Prof. Dr. Sina Saravi (Automatic Control Group, RAT).
Focus: signal and information processing, SLAM, neural networks (CNNs, SNNs), event-based vision.

B.Tech. Electronics and Communication Engineering 2015 – 2019
Madan Mohan Malaviya University of Technology, India
Bachelor's thesis: self-balancing object-tracking robot (PID-controlled inverted pendulum, IMU fusion, OpenCV tracking).

Research experience

Master's thesis research completed Oct 2026
Automatic Control Group (RAT), Paderborn University

- Core: replaced the CNN patch selector of the learned event-based visual odometry pipeline DEVO with spiking neural networks trained with surrogate gradients; compared two temporal encodings (replayed voxel grid, time-surface frames) against CNN baselines in 150 visual-odometry runs.
- On MVSEC, the spiking selector achieved about 2× lower absolute trajectory error and 10× smaller spread than the CNN baselines; on held-out TartanEvents sequences both viable spiking variants beat both CNNs.
- Showed with a same-data CNN control that the remaining gap on fast drone-racing motion (UZH-FPV) comes from the training data, not from spiking computation.
- Generated and publicly released **TartanEvents**, a 1.3 TB simulated event-camera training dataset.
- Extension: built **SuperEventSLAM**, the first graph-based SLAM back-end for DEVO (keyframes, SuperEvent features, three-stage loop verification, g2o pose-graph optimization). The pipeline runs end to end at 36.7 FPS; loop verification is not yet reliable, so drift reduction is future work.

Student assistant, humanoid robot learning Sep 2026 – present
Stratosfare – Fraunhofer IEM, Paderborn

- Implemented NVIDIA's GR00T end-to-end workflow on a real Unitree G1: teleoperated demonstrations, LeRobot dataset export, fine-tuning of the GR00T N1.7 vision-language-action model, and deployment via NVIDIA Jetson AGX Thor.
- Collecting whole-body VLA training data with NVLabs GR00T-WholeBodyControl (GEAR-SONIC, PICO VR teleoperation).
- Demonstrated the trained G1 picking an object and placing it in a container at a 3-day workshop and guided attendees through teleoperation.

Master's project: autonomous mapping and SLAM optimization
Paderborn University
LOAM-based lidar SLAM with IMU pose estimation; templated C++ architecture to benchmark pose-graph back-ends (g2o, SE-Sync) in ROS.

Publications, datasets and software

- E. Bajpai. *Exploring Neuromorphic Front-Ends for Event-Based SLAM.* Master's thesis, Paderborn University, 2026.
- Conference paper based on the thesis: manuscript in preparation.

- **TartanEvents** (dataset, 1.3 TB): Hugging Face: upb-rat-ekansh/TartanEventsLeft_wo_upsampling
- **pangolin-python-viz** (software): modular Python visualisation library on Pangolin for SLAM: github.com/ekansh-bajpai/pangolin-python-viz
- **devo-vo** (software): portable, self-contained subset of DEVO for event-based visual odometry: github.com/ekansh-bajpai/devo-vo
- **snn-devo** (spiking patch selector for DEVO) and **supereventslam** (SLAM back-end): private repositories, access on request.

Teaching

Student assistant, Nonlinear Control of Autonomous and Robotic Systems Apr 2026 – Sep 2026
Automatic Control Group (RAT), Paderborn University
Built the ROS 2 / Gazebo quadrotor framework and tutorials (PID, feedback linearization, sliding mode control, minimum-snap trajectories); supported Lyapunov-based stability analysis; graded student term projects. Course repository.

Industry experience

Student assistant, AI/ML and software 2024 – present
Fraunhofer IEM, Paderborn
LLM-based knowledge management and compliance platforms (retrieval-augmented generation, FastAPI back-ends).

Working student, front-end developer Feb 2023 – Sep 2023
NFDI4Culture (National Research Data Infrastructure), Paderborn University

Robotics engineering intern Aug 2022
Cellgo GmbH
Embedded I2C/SPI communication for a carrier robot (ESP32, Arduino).

Associate software developer Mar 2021 – Oct 2021
Sabre Corporation

Software engineer Jun 2019 – Feb 2021
HummingWave Technologies

Skills

Perception	Event cameras, spiking neural networks (snnTorch), visual odometry, SLAM, pose-graph optimization, bundle adjustment, OpenCV
Robot learning	Imitation learning, VLA fine-tuning (GR00T, LeRobot), teleoperation, whole-body control, Unitree G1
Control	Nonlinear control, sliding mode, feedback linearization, Lyapunov stability, PID, trajectory generation
Programming	Python, C++, PyTorch, ROS 2, Eigen, g2o, Gazebo, Git, Linux

References available on request.